Digital Protective Relay Simulator

Sponsor: Self, No Sponsor (Senior Design Student)

Background

Protective relays are used in electrical power systems to detect abnormal conditions and protect equipment from damage. Commercial protective relays, such as those manufactured by Schweitzer Engineering Laboratories (SEL) and Siemens provide these functions but can be expensive for small facilities and university laboratories.

This project will develop a **low-cost, microcontroller-based digital protective relay prototype** that demonstrates the basic operation of a commercial protective relay. The system will measure voltage and current, identify abnormal electrical conditions, and provide a trip signal when programmed protection limits are exceeded.

The project will provide hands-on experience with **power system protection, electronics, embedded programming, and digital signal processing**.

Project Needs

1. Measure low-voltage AC voltage and current signals.
2. Process the signals using a microcontroller.
3. Detect abnormal electrical conditions.
4. Apply basic protective relay functions.
5. Provide a trip signal when a protection limit is exceeded.

Project Scope

1. **Signal Measurement:** Design circuits to safely measure and scale low-voltage AC voltage and current signals for the microcontroller.
2. **Microcontroller and Programming:** Program a microcontroller to sample the signals, calculate RMS voltage and current, and determine when protection limits have been exceeded.
3. **Protection Functions:** Implement the following basic ANSI protection functions:
    - **ANSI 50 – Instantaneous Overcurrent**
    - **ANSI 51 – Time Overcurrent**
    - **ANSI 27 – Undervoltage**
    - **ANSI 59 – Overvoltage**
4. **User Interface:** Create a simple interface to display measured voltage and current, protection settings, and trip status.

5. **Trip Output:** Use a digital output to activate an external relay or indicator when a protection function operates.
6. **Testing:** Test the prototype using safe, low-voltage signals from laboratory equipment. The system will not be connected directly to utility-level voltages.

ANSI 81 frequency protection may be explored as a **stretch goal** if the primary functions are completed ahead of schedule.

Technical Considerations

1. Safely measuring and scaling voltage and current signals.
2. Accurately calculating RMS voltage and current.
3. Implementing appropriate timing for the ANSI 51 time-overcurrent function.
4. Programming reliable protection and trip logic.
5. Integrating the analog circuitry, microcontroller, user interface, and trip output.
6. Testing the system under different simulated fault conditions.

Deliverables

1. **Working Prototype:** A functional digital protective relay prototype capable of measuring voltage and current and producing a trip signal.
2. **Protection Functions:** Working demonstrations of ANSI 50, 51, 27, and 59 protection.
3. **User Interface:** A simple interface for displaying measurements, protection settings, and trip status.
4. **Testing Results:** Experimental results showing that the prototype responds appropriately to simulated overcurrent and abnormal voltage conditions.
5. **Technical Documentation:** Documentation describing the hardware design, software, protection functions, testing procedures, and results.
6. **Final Report and Presentation:** A final report and demonstration of the completed system.

Assumptions and Dependencies

1. Access to a microcontroller and basic electronic components.
2. Access to laboratory equipment such as an oscilloscope and function generator.
3. Testing will be performed using safe, low-voltage signals.
4. The project will prioritize the core protection functions before optional features are attempted.
5. ANSI 81 frequency protection will only be pursued if sufficient time remains after completing the primary project requirements.

Why Choose This Project?

- **Real-World Application:** Demonstrates protection functions commonly used in the power industry.
- **Manageable Scope:** Focuses on four fundamental protection functions while allowing additional features to be added if time permits.
- **Hands-On Experience:** Combines power systems, electronics, embedded programming, and digital signal processing.
- **Low Cost:** Provides an affordable platform for demonstrating basic protective relay concepts.
- **Industry Relevance:** Gives students practical experience with concepts used in commercial digital protective relays.